EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Subject: Low Code Application Development using Zoho Creator

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Aim

To implement and explore the security, access-control, data-sharing and governance features of the Event Registration application in Zoho Creator.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Event Registration application. The application is used for managing event registrations, and it serves as the practical environment for exploring the security and governance features described in this report.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram that shows the security and governance structure of the Event Registration application. It is not a screenshot of the Zoho Creator interface.

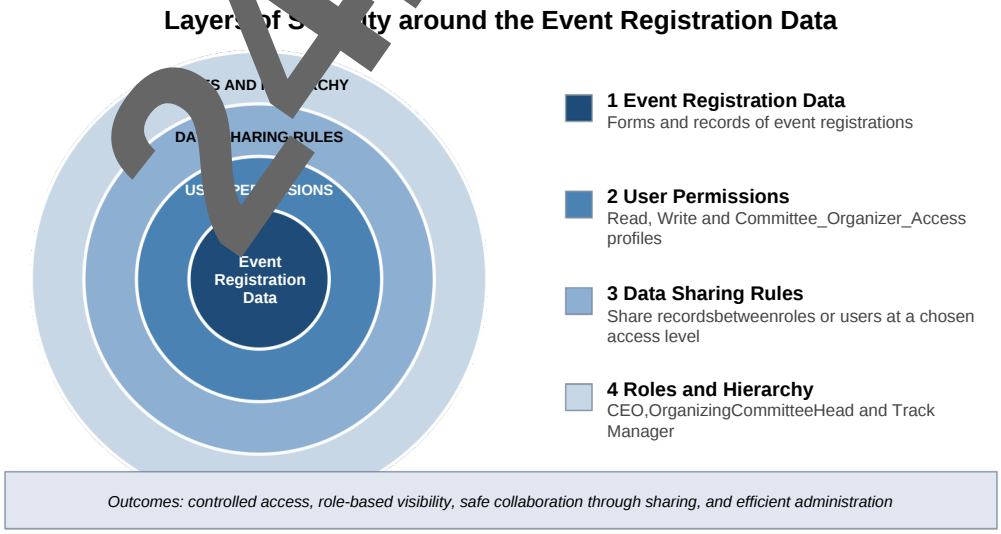


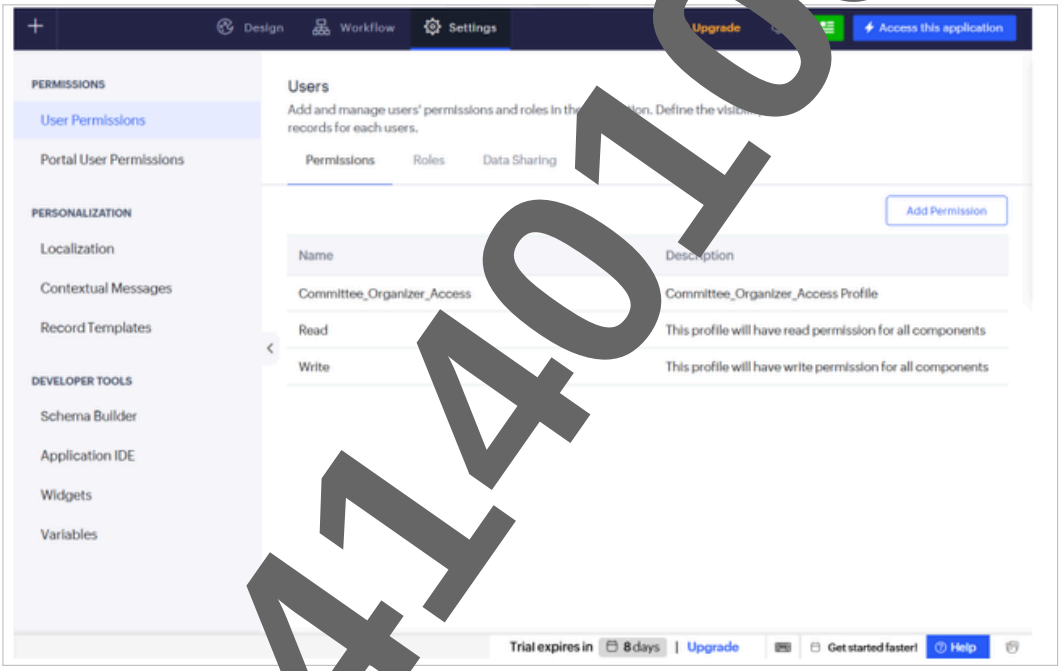
Figure 1: Security & Governance Layers (Conceptual Diagram)

The diagram shows the registration data of the application at the core, protected by successive layers. User permissions (Read, Write and Committee_Organizer_Access) define what a user may do. Data sharing rules extend the visibility of records between roles or users at a chosen level of access. The role hierarchy of CEO, Organizing Committee Head and Track Manager forms the outermost layer and influences which records each role can see. The key outcomes are controlled access to data, role-based visibility, safe collaboration through data sharing, and efficient administration.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application listed three permissions:

Name	Description (as displayed)
Committee_Organizer_Access	Committee_Organizer_Access Pr e
Read	This profile will have read per sion mponents
Write	This profile will have write permiss for al mponents



re 2: Permissions in the Event Registration application

Read permission a ws user to view the components and records of the application without changing them, whereas Wr perm sion llows the user to make changes, such as adding or modifying records. Committee_Organizer_Ac s a permission profile named for the committee organizers of the event application; it is listed alon e the Read and Write profiles and can be assigned to users who organise events.

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Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the Event Registration application the hierarchy is displayed with CEO at the top, Organizing Committee Head below it and Track Manager below that. The options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

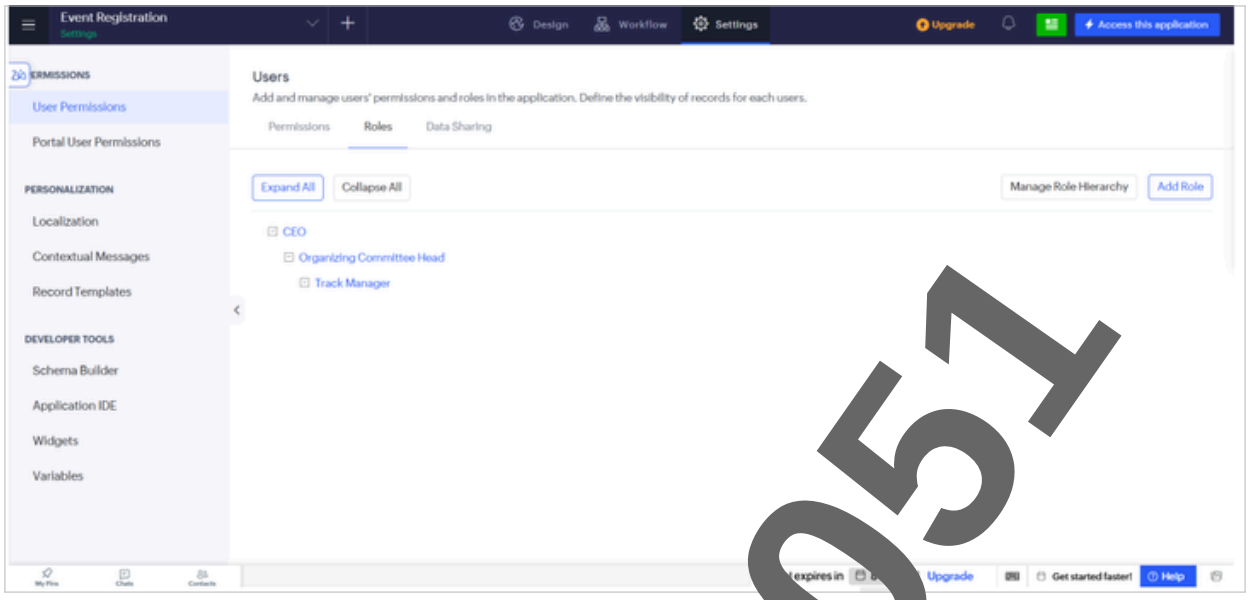


Figure 3: Roles and Role Hierarchy in the Event Registration Application

The roles shown in the hierarchy are summarised below.

Role	Reports To	Position in Hierarchy
CEO	–	Top level of the hierarchy
Organizing Committee Head	CEO	Second level
Track Manager	Organizing Committee Head	Third level (lowest role shown)

Arranging roles in this way gives the application a clear reporting structure. The position of a role in the hierarchy influences which records its members can see, and further access can be given to a specific role through data sharing rules, as described in the next section.

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Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. A data sharing rule was configured with the following details:

Parameter	Value
Rule Name	Event_Registration_Form
Form	Event Registration
Records Shared From	Roles – Organizing Committee Head
Records Shared To	Roles – Track Manager
Level of Access	Read Only

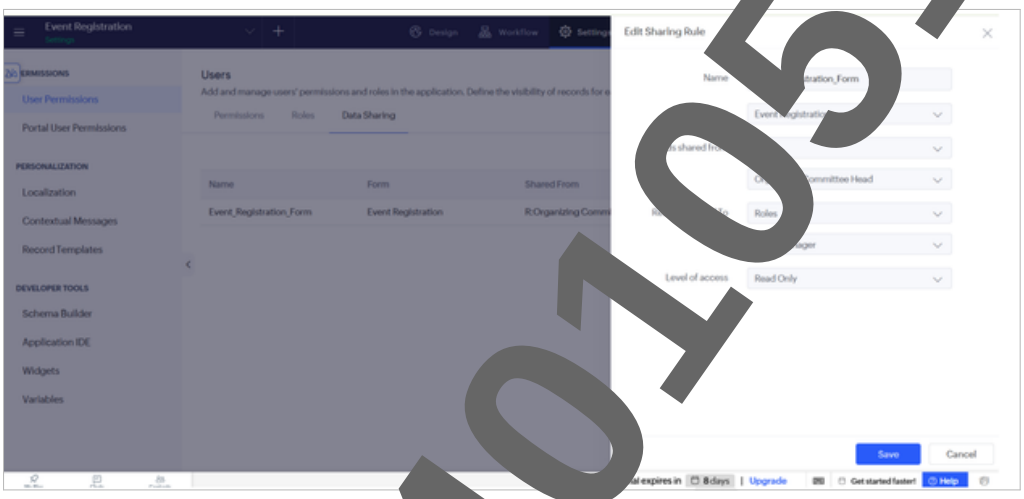


Figure 4: Edit Sharing Rule p… …g the Event_Registration_Form rule

Figure 4 shows the saved rule in edit m…de; t… …le also appears in the Data Sharing list in the background with its name, form and the sh… …rom role. While creating a rule, the New Share Rule panel offers three choices under Re… …hared from (Figure 5).

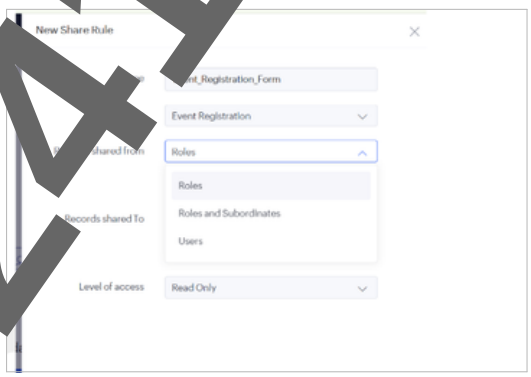


Figure 5: New Share Rule panel – options for Records shared from

The three options are Roles, Roles and Subordinates, and Users. Roles shares the records of a single selected role, Roles and Subordinates shares the records of a role together with the roles beneath it in the hierarchy, and Users shares the records of selected individual users. In this experiment Roles was selected, with the Organizing Committee Head role as the source. With Read Only access, members of the Track Manager role can view the shared records of the Event Registration form but cannot modify them.

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Security Features and Governance Controls

Figure 6 is a conceptual diagram that classifies the security and governance controls of Zoho Creator as they apply to the Event Registration application. It is not a screenshot, and it is not proof that every listed feature was configured.

Classification of Security and Governance Controls

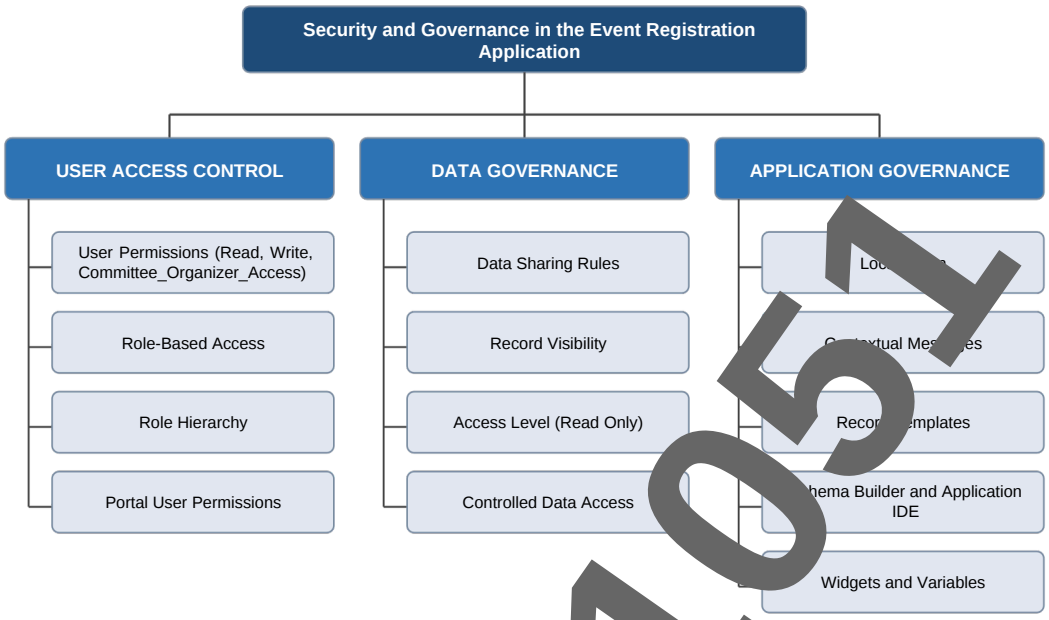


Figure 6: Classification of Security and Governance Controls (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read, Write, and Committee_Organizer_Access), role-based and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, access level, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and the role hierarchy were examined, and a data sharing rule was configured. The other items in the diagram, such as portal user permissions, localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

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Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was not configured in this experiment, and no portal permissions were set for the Event Registration application.

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram that combines the role hierarchy of the application with the data sharing rule that was configured. On the left, the roles CEO, Organizing Committee Head and Track Manager are shown in reporting order. On the right, the rule Event_Registration_Form is traced from the Event Registration form, through the role it is shared from, to the role it is shared to and the level of access given.

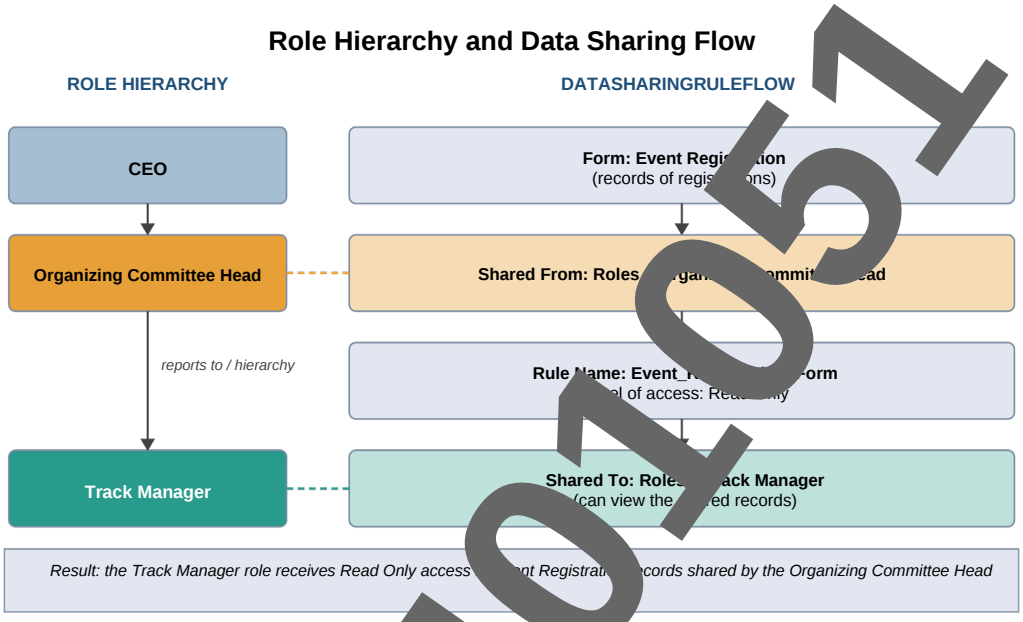


Figure 7: Role Hierarchy and Data Sharing Flow (Conceptual Diagram)

- The roles of the application are arranged in a hierarchy: CEO, Organizing Committee Head and Track Manager.
- The data sharing rule Event_Registration_Form is defined for the Event Registration form.
- Records are shared from the Organizing Committee Head role to the Track Manager role.
- The level of access is Read Only, so the Track Manager role can view the shared records but cannot modify them.

Governance Observations

- Predefined Read and Write permissions were available in the application, together with the Committee_Organizer profile.
- A data sharing rule named Event_Registration_Form was configured for the Event Registration form.
- The rule shares records from the Organizing Committee Head role to the Track Manager role with Read Only access.
- Portal User Permissions was not configured in this experiment.
- The practical showed how access can be limited to what a role needs by choosing a suitable level of access.

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Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the Read, Write and Committee_Organizer_Access permissions of the Event Registration application were examined, the role hierarchy of CEO, Organizing Committee Head and Track Manager was viewed, and a data sharing rule named Event_Registration_Form was configured to share the records of the Event Registration form from the Organizing Committee Head role to the Track Manager role with Read Only access.

The experiment shows how permissions, roles and data sharing work together so that each role sees the records it needs and no more. Choosing a suitable level of access, such as Read Only, allows records to be shared for collaboration without allowing them to be changed. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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